

Ph.D. Qualifying Exam 2018: Probability

August 20, 2018

10 AM to 2 PM

Rules: This is a four hour exam. You may use ONE page (front and back, 8.5 by 11 inch paper) HANDWRITTEN “cheat sheet”. No other references are allowed. No use of cell phones, calculators, computers, or other electronic devices is permitted for any reason during the time while the exam is being administered, including any time during which you leave the exam room for any reason. **Show your work and justify your answers clearly. If you use known results (e.g., results stated/proven in class), please state them clearly.** Please put your name or initials on every page. Good luck!

Problem 1

Name _____

Let X_1, X_2 be i.i.d. random variables with a Uniform distribution on $[0, 1]$. Further let

$$T = \max\{X_1, \bar{X}\},$$

where $\bar{X}$ is the sample mean of X_1 and X_2 .

- (a) Find the cumulative distribution function of T .
- (b) Find $E[T]$.
- (c) Find $E[X_1|T = t]$ for $0 \leq t \leq 1$.
- (d) Find $Var[X_1|T = t]$ for $0 \leq t \leq 1$.

Problem 2

Name _____

Let $\mathcal{X}$ be a finite set and let $P(\cdot, \cdot)$ be an irreducible and aperiodic transition matrix on $\mathcal{X}$, with stationary distribution π .

Random mapping representation. Throughout the problem, we assume that there exists a random variable Z valued in some set Λ and function $f : \mathcal{X} \times \Lambda \rightarrow \mathcal{X}$ such that for any $x, y \in \mathcal{X}$ we have $\mathbf{P}(f(x, Z) = y) = P(x, y)$.

Sequence $(Z_t)_{t \in \mathbb{Z}}$ of iid random variables. Throughout the problem, let $(Z_t)_{t \in \mathbb{Z}}$ be a countable sequence of iid copies of Z , indexed by the set of integers $\mathbb{Z}$. Hence for any $t \in \mathbb{Z}$ we have $\mathbf{P}(f(x, Z_t) = y) = P(x, y)$ for all $x, y \in \mathcal{X}$.

Assumption (C). The Markov Chain and the above random mapping representation is such that, for the grand coupling defined by

$$\forall x \in \mathcal{X}, \quad X_0^x = x \quad \text{and} \quad \forall t \geq 1, \quad X_t^x = f(X_{t-1}^x, Z_t),$$

the coalescence time τ_c is finite with probability one, where

$$\tau_c = \begin{cases} +\infty & \text{if the set } \{t \geq 0 : X_t^x = X_t^y \text{ for all } x, y \in \mathcal{X}\} \text{ is empty,} \\ \min\{t \geq 0 : X_t^x = X_t^y \text{ for all } x, y \in \mathcal{X}\} & \text{if the set is non-empty.} \end{cases}$$

Part 0

0. Give an example of a random mapping representation of some aperiodic and irreducible Markov Chain such that Assumption (C) does not hold, that is, an example with $\mathbf{P}(\tau_c = +\infty) > 0$.

In the rest of the exam, we assume that Assumption (C) always holds.

Part I

1. Give an example of a random mapping representation of some aperiodic and irreducible Markov Chain on $\mathcal{X} = \{0, 1, 2\}$ such that the random variable $X_{\tau_c}^{x_0}$ for $x_0 \in \mathcal{X}$ is NOT distributed according to π .

Part II

For any $t \in \mathbb{Z}$, define the random function

$$\begin{aligned} f_t : \mathcal{X} &\rightarrow \mathcal{X} \\ x &\rightarrow f_t(x) = f(x, Z_t), \end{aligned}$$

as well as for any $s, t \in \mathbb{Z}$ with $s < t$, the random function

$$F_s^t = f_{t-1} \circ f_{t-2} \circ f_{t-3} \circ \dots \circ f_{s+1} \circ f_s.$$

For instance, we have

- $F_t^{t+1}(x) = f_t(x) = f(x, Z_t)$ for any $x \in \mathcal{X}, t \in \mathbb{Z}$.
- $F_{-3}^0(x) = (f_{-1} \circ f_{-2} \circ f_{-3})(x) = f(f(f(x, Z_{-3}), Z_{-2}), Z_{-1})$ for any $x \in \mathcal{X}$,
- $F_{-t}^0 \circ F_{-2t}^{-t} = F_{-2t}^0$ for any $t \geq 0$,
- The grand coupling above is then given by $X_t^x = F_1^{t+1}(x)$ for $t \geq 1$, and $x \in \mathcal{X}$.

2. Prove that for fixed $t > 0, x_0 \in \mathcal{X}, y \in \mathcal{X}$,

- $\mathbf{P}(F_{-t}^0(x_0) = y) = \mathbf{P}(F_1^{t+1}(x_0) = y)$.
- $\mathbf{P}(\forall x \in \mathcal{X}, F_{-t}^0(x) = y) = \mathbf{P}(\forall x \in \mathcal{X}, F_1^{t+1}(x) = y)$.

3. We say that a function $F : \mathcal{X} \rightarrow \mathcal{X}$ is constant if and only if the set $\{F(x), x \in \mathcal{X}\}$ has exactly one element.

Define $\mathcal{M} = \min\{t > 0 : F_{-t}^0 \text{ is constant}\}$. Prove that $\mathcal{M}$ is finite with probability one under Assumption (C).

4. Prove that $P(F_{-\mathcal{M}}^0(x) = F_{-k-\mathcal{M}}^0(y)) = 1$ for any $x, y \in \mathcal{X}$ and any integer $k \geq 0$, where $\mathcal{M}$ is as defined above.

5. *Most important and difficult question of the problem. Make sure to attend this question and be clear and rigorous in your answer.*

Prove that $F_{-\mathcal{M}}^0(x_0)$ has distribution π for any $x_0 \in \mathcal{X}$.

Hint: you may prove and use the following fact: if a random variable $\hat{X}$ independent of Z is such that $f(\hat{X}, Z)$ has the same distribution as $\hat{X}$, then $\hat{X}$ has distribution π .

6. Deduce from the previous question an algorithm that outputs a random variable distributed according to π , by sequentially generating iid random variables $Z_{-1}, Z_{-2}, Z_{-3}, \dots$

Part III

Consider now the following algorithm.

- ALGORITHM 2:

- Set $t = 1$.
 - Generate $Z_{-1}, Z_{-2}, \dots, Z_{-t}$ iid copies of Z independently of all previous iterations of the algorithm
 - If F_{-t}^0 is constant, then output $F_{-t}^0(x_0)$ for some $x_0 \in \mathcal{X}$ and stop the algorithm.
 - If F_{-t}^0 is not constant, then throw away $Z_{-1}, Z_{-2}, \dots, Z_{-t}$, increase t by one, i.e., set $t \leftarrow t + 1$, and go to step b.
7. By studying the Markov Chain defined on $\mathcal{X} = \{0, 1, 2\}$ by the following graph, with the random mapping representation $f(i, z) = \max(0, \min(i + z, 2))$ with $\mathbf{P}(Z = 1) = \mathbf{P}(Z = -1) = 1/2$, find the stationary distribution π and show that Algorithm 2 does not output a random variable distributed according to π . (*Hint: you may, for instance, show that if $\hat{Y}$ is the random variable output by Algorithm 2 then $\mathbf{P}(\hat{Y} \in \{0, 2\})$ is too large.*)

